SERIAL COMMUNICATIONS

I²C

With $\overline{CS}$ tied high to $V_{DD\ I/O}$, the ADXL345 is in I²C mode, requiring a simple 2-wire connection, as shown in Figure 40. The ADXL345 conforms to the *UM10204 I²C-Bus Specification and User Manual*, Rev. 03—19 June 2007, available from NXP Semiconductors. It supports standard (100 kHz) and fast (400 kHz) data transfer modes if the bus parameters given in Table 11 and Table 12 are met. Single- or multiple-byte reads/writes are supported, as shown in Figure 41. With the ALT ADDRESS pin high, the 7-bit I²C address for the device is 0x1D, followed by the R/$\overline{W}$ bit. This translates to 0x3A for a write and 0x3B for a read. An alternate I²C address of 0x53 (followed by the R/$\overline{W}$ bit) can be chosen by grounding the SDO/ALT ADDRESS pin (Pin 12). This translates to 0xA6 for a write and 0xA7 for a read.

There are no internal pull-up or pull-down resistors for any unused pins; therefore, there is no known state or default state for the $\overline{CS}$ or ALT ADDRESS pin if left floating or unconnected. It is required that the $\overline{CS}$ pin be connected to $V_{DD\ I/O}$ and that the ALT ADDRESS pin be connected to either $V_{DD\ I/O}$ or GND when using I²C.

Due to communication speed limitations, the maximum output data rate when using 400 kHz I²C is 800 Hz and scales linearly with a change in the I²C communication speed. For example, using I²C at 100 kHz would limit the maximum ODR to 200 Hz. Operation at an output data rate above the recommended maximum may result in undesirable effect on the acceleration data, including missing samples or additional noise.

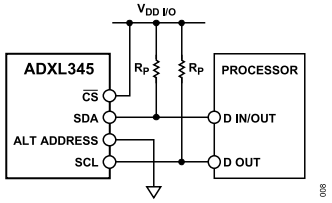


Figure 40. I²C Connection Diagram (Address 0x53)

If other devices are connected to the same I²C bus, the nominal operating voltage level of these other devices cannot exceed $V_{DD\ I/O}$ by more than 0.3 V. External pull-up resistors, R_P, are necessary for proper I²C operation. Refer to the *UM10204 I²C-Bus Specification and User Manual*, Rev. 03—19 June 2007, when selecting pull-up resistor values to ensure proper operation.

Table 11. I²C Digital Input/Output

Parameter	Test Conditions	Limit[1] Min	Limit[1] Max	Unit
Digital Input				
Low Level Input Voltage (V_{IL})			0.3 × $V_{DD\ I/O}$	V
High Level Input Voltage (V_{IH})		0.7 × $V_{DD\ I/O}$		V
Low Level Input Current (I_{IL})	V_{IN} = $V_{DD\ I/O}$		0.1	µA
High Level Input Current (I_{IH})	V_{IN} = 0 V	−0.1		µA
Digital Output				
Low Level Output Voltage (V_{OL})	$V_{DD\ I/O}$ < 2 V, I_{OL} = 3 mA		0.2 × $V_{DD\ I/O}$	V
	$V_{DD\ I/O}$ ≥ 2 V, I_{OL} = 3 mA		400	mV
Low Level Output Current (I_{OL})	V_{OL} = $V_{OL,\ max}$	3		mA
Pin Capacitance	f_{IN} = 1 MHz, V_{IN} = 2.5 V		8	pF

[1] Limits based on characterization results; not production tested.

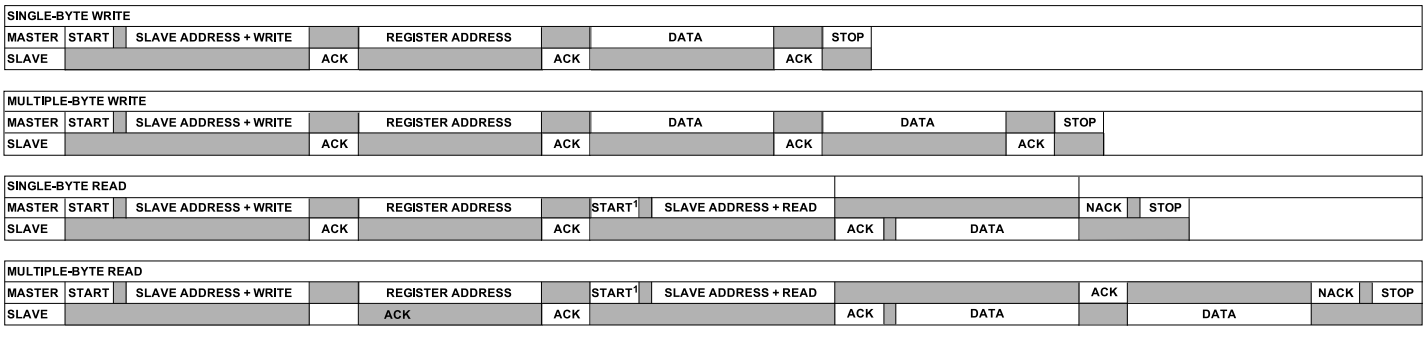


Figure 41. I²C Device Addressing

SERIAL COMMUNICATIONS

Table 12. I²C Timing (T_A = 25°C, V_S = 2.5 V, $V_{DD\ I/O}$ = 1.8 V)

	Limit[1, 2]			
Parameter	**Min**	**Max**	**Unit**	**Description**
f_{SCL}		400	kHz	SCL clock frequency
t_1	2.5		µs	SCL cycle time
t_2	0.6		µs	t_{HIGH}, SCL high time
t_3	1.3		µs	t_{LOW}, SCL low time
t_4	0.6		µs	$t_{HD,\ STA}$, start/repeated start condition hold time
t_5	100		ns	$t_{SU,\ DAT}$, data setup time
t_6[3, 4, 5, 6]	0	0.9	µs	$t_{HD,\ DAT}$, data hold time
t_7	0.6		µs	$t_{SU,\ STA}$, setup time for repeated start
t_8	0.6		µs	$t_{SU,\ STO}$, stop condition setup time
t_9	1.3		µs	t_{BUF}, bus-free time between a stop condition and a start condition
t_{10}		300	ns	t_R, rise time of both SCL and SDA when receiving
	0		ns	t_R, rise time of both SCL and SDA when receiving or transmitting
t_{11}		300	ns	t_F, fall time of SDA when receiving
		250	ns	t_F, fall time of both SCL and SDA when transmitting
C_b		400	pF	Capacitive load for each bus line

1 Limits based on characterization results, with f_{SCL} = 400 kHz and a 3 mA sink current; not production tested.

2 All values referred to the V_{IH} and the V_{IL} levels given in Table 11.

3 t_6 is the data hold time that is measured from the falling edge of SCL. It applies to data in transmission and acknowledge.

4 A transmitting device must internally provide an output hold time of at least 300 ns for the SDA signal (with respect to $V_{IH(min)}$ of the SCL signal) to bridge the undefined region of the falling edge of SCL.

5 The maximum t_6 value must be met only if the device does not stretch the low period (t_3) of the SCL signal.

6 The maximum value for t_6 is a function of the clock low time (t_3), the clock rise time (t_{10}), and the minimum data setup time ($t_{5(min)}$). This value is calculated as $t_{6(max)} = t_3 - t_{10} - t_{5(min)}$.

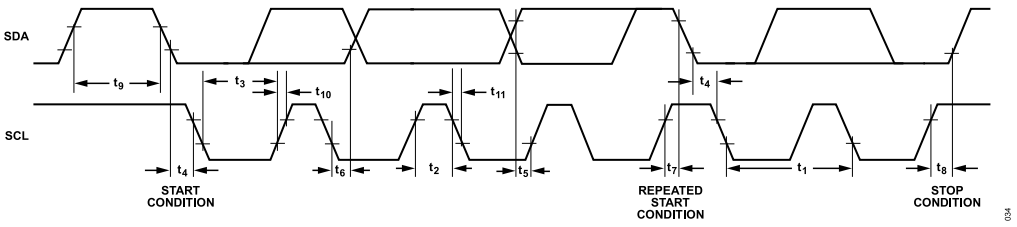


Figure 42. I²C Timing Diagram

INTERRUPTS

The ADXL345 provides two output pins for driving interrupts: INT1 and INT2. Both interrupt pins are push-pull, low impedance pins with output specifications shown in Table 13. The default configuration of the interrupt pins is active high. This can be changed to active low by setting the INT_INVERT bit in the DATA_FORMAT (Address 0x31) register. All functions can be used simultaneously, with the only limiting feature being that some functions may need to share interrupt pins.

Interrupts are enabled by setting the appropriate bit in the INT_ENABLE register (Address 0x2E) and are mapped to either the INT1 pin or the INT2 pin based on the contents of the INT_MAP register (Address 0x2F). When initially configuring the interrupt pins, it is recommended that the functions and interrupt mapping be done before enabling the interrupts. When changing the configuration of an interrupt, it is recommended that the interrupt be disabled first, by clearing the bit corresponding to that function in the INT_ENABLE register, and then the function be reconfigured before enabling the interrupt again. Configuration of the functions while the interrupts are disabled helps to prevent the accidental generation of an interrupt before desired.

The interrupt functions are latched and cleared by either reading the data registers (Address 0x32 to Address 0x37) until the interrupt condition is no longer valid for the data-related interrupts or by reading the INT_SOURCE register (Address 0x30) for the remaining interrupts. This section describes the interrupts that can be set in the INT_ENABLE register and monitored in the INT_SOURCE register.

Table 13. Interrupt Pin Digital Output

		Limit[1]		
Parameter	**Test Conditions**	**Min**	**Max**	**Unit**
Digital Output				
Low Level Output Voltage (V_{OL})	I_{OL} = 300 µA		0.2 × $V_{DD\ I/O}$	V
High Level Output Voltage (V_{OH})	I_{OH} = −150 µA	0.8 × $V_{DD\ I/O}$		V
Low Level Output Current (I_{OL})	V_{OL} = $V_{OL,\ max}$	300		µA
High Level Output Current (I_{OH})	V_{OH} = $V_{OH,\ min}$		−150	µA
Pin Capacitance	f_{IN} = 1 MHz, V_{IN} = 2.5 V		8	pF
Rise/Fall Time				
Rise Time (t_R)[2]	C_{LOAD} = 150 pF		210	ns
Fall Time (t_F)[3]	C_{LOAD} = 150 pF		150	ns

[1] Limits based on characterization results, not production tested.

[2] Rise time is measured as the transition time from $V_{OL,\ max}$ to $V_{OH,\ min}$ of the interrupt pin.

[3] Fall time is measured as the transition time from $V_{OH,\ min}$ to $V_{OL,\ max}$ of the interrupt pin.